

Module 2 Lab Reflective Journal

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Course: ITAI 1371

Assignment: Module 2 Lab Exercise

Module 2 helped me understand how pandas, NumPy, and matplotlib can work together during the early stages of a machine learning project. This reflection also helped me identify the areas where I need more practice and what I can improve in future labs and projects.

One of the main concepts I learned in this lab was how pandas organize a dataset into a DataFrame. The DataFrame presented the information through rows and columns, making the data easier to read and analyze with Python. NumPy also helped me understand how numerical information can be summarized efficiently. For example, it allowed me to calculate the mean and standard deviation of the sepal length.

In the visualizations, I noticed several patterns while comparing the three flower species:

- Setosa generally had a wider sepal than versicolor and virginica.
- Versicolor and virginica had similar sepal lengths and widths, causing their points to overlap in the scatter plot.
- Setosa had the shortest sepal length of the three species.

These patterns were easier to recognize in the graphs than in the numerical table. This showed me that visualizations can make comparisons and relationships within a dataset easier to understand.

In this lab, I also learned that a notebook contains both code cells and Markdown cells. At first, I expected every cell to contain code, but I learned that Markdown is important for explaining and documenting an analysis. It helps me to understand the purpose of the code and the meaning of its results.

Overall, this lab gave me an introduction to how datasets and DataFrames function in Python. I learned how a DataFrame can organize information into rows and columns that Python can process and analyze. I still need more practice with the different commands and functions, but this lab gave me a stronger foundation for future machine learning assignments and projects.